



The Federal University of Technology, Akure

School of Engineering and Engineering Technology

Department of Computer Engineering

Second Semester 2023/2024 Academic Session Examination

COURSE – CPE510: Computer Security Techniques

CREDIT UNITS: 2

DATE: 24/09/2024

Instruction: Attempt ANY FOUR Questions

Time Allowed: $2\frac{1}{2}$ Hours



- 1a. Consider the finite-state machine in Figure 1. It consists of four states and five transitions. The security policy partitions the states into a set of authorized states $A = \{s_1, s_2\}$ and a set of unauthorized states $UA = \{s_3, s_4\}$. Is the system secure or otherwise? Justify your answer. Suppose that edge t_3 went from s_1 to s_4 .

Would the resulting system be secure or insecure?

2 marks

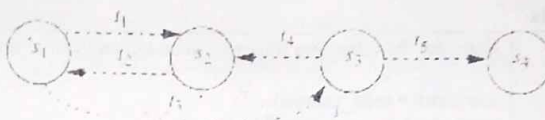


Figure 1: Simple Finite-state Machine. In this example, the authorized states are s_1 and s_2 .

- 1b. With the aid of a diagram, explain the overall architecture for a distributed intrusion detection system consisting of its three (3) main components.

9 marks

- 1c. List and explain two (2) typical phases of operation of a virus or worm during its lifetime?

4 marks

- 2a. With the aid of diagram, explain the six (6) steps required to generate the ciphertext from a plaintext using the Data Encryption Standard (DES).

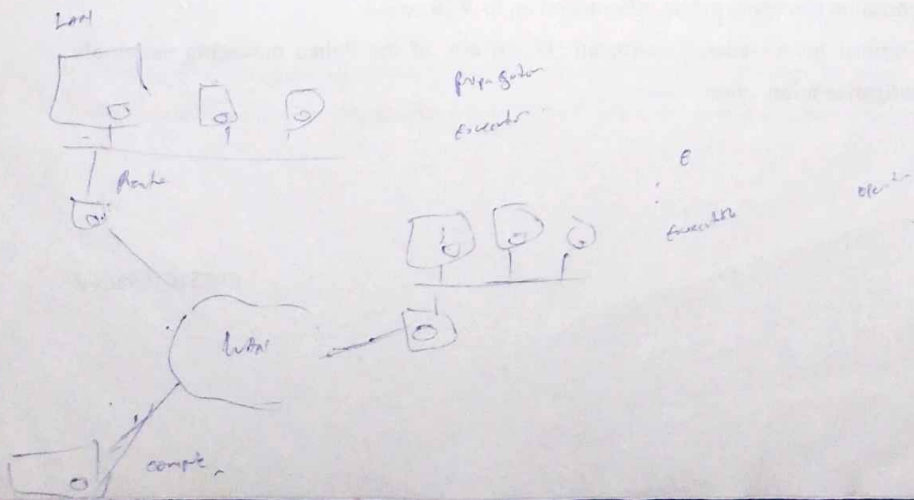
6 marks

- 2b. What is the purpose of classic security models? Describe the Bell-LaPadula security model using a well labelled diagram.

4 marks

- 2c. The index of Coincidence (IC) is defined as "the probability that two randomly chosen letters from the ciphertext will be the same." Derive the formula for IC and find IC for a randomly generated text.

5 marks



3a. Explain the following terminologies in relation to computer security with one scenario each.

8 marks

i. Social Engineering

ii. Denial of Service *Man in the middle*

iii. Trojan Horse

iv. Worm *Phishing*

3b. Consider a telephone switching system that routes calls through a switching network based on the telephone number requested by the caller. Give examples of confidentiality, integrity, and availability requirements associated with the system. In each case, indicate the degree of importance of the requirement as either high or low.

5marks

3c. Consider the code fragments given in Table 1, what type of malicious software are they?

2 marks

Table 1: Code Fragments

<i>legitimate code</i> if data is Friday the 13th; crash_computer(); <i>legitimate code</i>	Consider the following fragment in an authentication program: username = read_username (); password = read_password (); if username is "133t h4ck0r" return ALLOW_LOGIN; if username and password are valid return ALLOW_LOGIN else return DENY_LOGIN
--	--

4a. For each of the following assets, assign a low, moderate, or high impact level for the loss of confidentiality, availability, and integrity, respectively. Justify your answers.

4 marks

- A commercial bank managing routine administrative information (not privacy-related information). *LMM*
- A power plant contains a SCADA (Supervisory Control and Data Acquisition) system controlling the distribution of electric power for a large military installation. The SCADA system contains both real-time *THH* sensor data and routine *MMU* administrative information. Assess the impact for the two data sets separately.
- A media organization managing public information on its Web server. *LWH*
- The Force Criminal Investigation Department (FCID) arm of the Police managing extremely sensitive investigative information. *HAM*

4b. Consider the plaintext of Martin Fowler's quote: "ANYFO OLCAN WRITE CODE? HATAC OMPUT ERCAN UNDER STAND". Calculate the plaintext's index of coincidence (IC) and comment on your answer.

6 marks

4c. What is Firewall? State four (4) advantages of implementing a Firewall in a private internal network.

5 marks

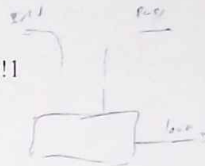
5a. Use the Vigenère algebraic formula to encipher the plaintext "A LIMERICK PACKS LAUGHS ANATOMICAL" using the key "BENCH".

7 marks

5b. Explain the suitability or unsuitability of the following passwords:

3 marks

- i. LND - 13CV
- ii. Mfmitm@(for "my favorite movie is tender mercies")!
- iii. Natalie1
- iv. Washington
- v. @tv9stove%
- vi. 12345678



5 marks

5c. Explain five (5) motives behind network and computer attacks.

6a. List and explain three (3) techniques used by firewalls to control access and enforce a security policy.

3 marks

6b. A widely used password security technique is the use of hashed password and a salt value. With the aid of suitable diagrams, explain the procedure involved in loading a new password and verifying it on UNIX variants as well as a number of other operating systems.

10 marks

6c. The question arises as to whether it is possible to develop a program that can analyze a piece of software to determine if it is a virus. Consider that we have a program D that is supposed to be able to do that. That is, for any program P , if we run $D(P)$, the result returned is TRUE (P is a virus) or FALSE (P is not a virus).

Now consider the following program:

2 marks

```

Program CV :=
{ ...
main-program :=
    { if D(CV) then goto next:
      else infect-executable;
    } next:
}
    
```



Handwritten notes:
 Go through from original virus...
 infect-executable...
 memory...
 program...
 loading data...
 instructions...

In the preceding program, infect-executable is a module that scans memory for executable programs and replicates itself in those programs. Determine if D can correctly decide whether CV is a virus.

Good luck



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B. ENG. DEGREE EXAMINATIONS. SECOND SEMESTER 2024/2025 SESSION

COURSE CODE: CPE510

COURSE TITLE: Computer Security Techniques

UNITS: 2

TIME ALLOWED: 150 minutes

Instruction: Attempt question ONE (1) and any other THREE (3) questions

- ✓ 1a. Break the ciphertext "KHOOR ZRUOG" by applying the character-based model to compute $\phi(l)$, which is the correlation of the frequency of each letter in the ciphertext with the character frequencies in English language (shown in Figure 1). State the three (3) most likely keys and the plaintext.

14 marks

$\phi = 0.14$

$e = 0.1$

a 0.080

h 0.060

n 0.070

t 0.090

b 0.015

i 0.065

o 0.080

u 0.030

c 0.030

j 0.005

p 0.020

v 0.010

d 0.040

k 0.005

q 0.002

w 0.015

0.14

e 0.130

l 0.035

r 0.065

x 0.005

f 0.020

m 0.030

s 0.060

y 0.020

g 0.015

z 0.002

Figure 1: Character Frequencies in English Language (Unigram Model of English)

- 1b. Consider the following scenarios and give examples of confidentiality, integrity and availability requirements associated with the system. In each case, indicate the degree of importance of the requirement.

6 marks

- An automated teller machine (ATM) in which users provide a personal identification number (PIN) and a card for account access.
- A telephone switching system that routes calls through a switching network based on the telephone number requested by the caller.

- 2a. Answer the following questions in relation to intruders.

5 marks

- List and briefly define three (3) classes of intruders.
- Explain how statistical anomaly detection and rule-based intrusion detection are used to detect different types of intruders.

- 2b. Using a simple virus logic, explain the general depiction of an executable virus structure. Why is the simple virus you have explained easily detectable and what way can this detection be thwarted.

10 marks



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2c. Describe five (5) goals an ideal password authentication scheme should achieve.

5 marks

3a. Explain with the aid of a well labelled diagram the most common distinction between an internal, an external firewall and a DMZ (demilitarized zone) network.

8 marks

3b. Use the Vigenère algebraic formula to encipher the plaintext "A LIMERICK PACKS LAUGHS ANATOMICAL" using the key "BENCH".

7 marks

3c. Explain the suitability or unsuitability of the following passwords:

3 marks

- LND - 13CV
- Mfmitm@(for "my favorite movie is tender mercies")!
- Natalie1
- Washington
- @tv9stove%
- 12345678

3d. Consider the finite-state machine in Figure 2. It consists of four states and five transitions. The security policy partitions the states into a set of authorized states $A = \{s_1, s_2\}$ and a set of unauthorized states $UA = \{s_3, s_4\}$. Is the system secure or otherwise? Justify your answer. Suppose that edge t_3 went from s_1 to s_4 .

Would the resulting system be secure or insecure?

2 marks

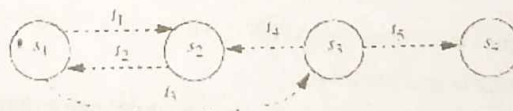


Figure 2: Simple Finite-state Machine. In this example, the authorized states are s_1 and s_2 .

4a. Explain the following terminologies in relation to computer security with one scenario each.

8 marks

- Social Engineering
- Denial of Service
- Trojan Horse
- Worm

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4b. Consider the plaintext of Martin Fowler's quote: "ANYFO OLCAN WRITE CODET HATAC OMPUT ER CAN UNDER STAND". Calculate the plaintext's index of coincidence (IC) and comment on your answer.

6 marks

4c. With the aid of diagram, explain the six (6) steps required to generate the ciphertext from a plaintext using the Data Encryption Standard (DES).

6 marks

5a. A widely used password security technique is the use of hashed password and a salt value. With the aid of suitable diagrams, explain the procedure involved in loading a new password and verifying it on UNIX variants as well as a number of other operating systems.

10 marks

5b. List and explain two (2) typical phases of operation of a virus or worm during its lifetime?

4 marks

5c. What is the purpose of classic security models? Describe the Clark-Wilson security model using a well labelled diagram.

6 marks

6a. List the three design goals of a Firewall.

3 marks

6b. The index of Coincidence (IC) is defined as "the probability that two randomly chosen letters from the ciphertext will be the same." Derive the formula for IC and find IC for a randomly generated text.

$$I_c = \frac{\sum_{i=1}^N (f_i - 1)}{N(N-1)}$$

5 marks

6c. Explain five (5) motives behind network and computer attacks.

5 marks

6d. Consider a desktop publishing system used to produce documents for various organizations.

3 marks

- Give an example of a type of publication for which confidentiality of the stored data is the most important requirement.
- Give an example of a type of publication in which data integrity is the most important requirement.
- Give an example in which system availability is the most important requirement.

6e. What is Firewall? State three (4) advantages of implementing a Firewall in a private internal network.

4 marks

Good luck